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Regularized quantile-based fuzzy regression with lasso penalty

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ABSTRACT

The lasso technique is a regularization method used in regression analysis, primarily to avoid overfitting and redundancy. This paper extends the application of lasso estimation to regression models in which predictors and responses are modeled using LR fuzzy numbers. To achieve this, a uniform quantile method is incorporated into the lasso technique, effectively mitigating the impact of large residuals while maintaining interpretable coefficients, all within a computationally efficient framework. By comparative analysis using simulation studies and real-world application examples we demonstrate the practical effectiveness and excellent performance of the introduced fuzzy regression model.

1. Introduction

Fuzzy regression models offer a versatile approach to regression analysis, particularly when dealing with uncertain or imprecise data, non-linear relationships, and the integration of qualitative information. By embracing uncertainty and flexibility, these models can provide more insightful and reliable insights into complex systems and phenomena. Against this backdrop, it is not surprising that fuzzy regression models have achieved great popularity and that there are numerous publications in this field [1]. To give the reader an impression of the diversity of fuzzy regression techniques, we briefly present various contributions in the following, limiting ourselves to the most recent developments.

Yoon et al. [2] proposed a fuzzy correlation coefficient and multiple fuzzy regression analysis incorporating a variable selection method, using defuzzification, fuzzy ordering, and distance-based approaches to enhance the analysis of ambiguous financial data. Chung [3] dealt with a fuzzy nonparametric regression method, which employs a convex nonparametric least squares approach to overcome the limitations of traditional fuzzy least squares models. Junior et al. [4] provided a comprehensive survey of deep fuzzy systems for regression applications, exploring the integration of deep learning and fuzzy systems to achieve both high accuracy and interpretability. Hesamian et al. [5] utilized the machine learning technique radial basis function networks for fuzzy regression modeling. Most recently, Hesamian et al. [6] considered a fuzzy nonlinear quantile-based regression model with crisp predictors and fuzzy responses. As the publications presented above are only a small excerpt from all the articles on new approaches in the area of fuzzy regression analysis, we refer the reader to Chukhrova and Johannssen [1].

When fitting a linear regression model $y_i = \beta_0 + \boldsymbol{x}_i^\top \boldsymbol{\beta} + \epsilon_i$, penalization methods are crucial in the identification of the relevant predictors [7,8]. Regularization methods can be used to select the predictor variables and simultaneously to estimate the model parameters. For instance, the *least absolute shrinkage and selection operator* (lasso, see Tibshirani [9]) serves the dual purpose of regularization and variable selection, aiming to improve both the predictive accuracy and interpretability of a regression model.

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The lasso technique improves the predictive accuracy by shrinking or setting some coefficients equal to zero by minimizing a proper target function such as

$$\sum_{i=1}^{n} d(y_i, \beta_0 + \boldsymbol{x}_i^\top \boldsymbol{\beta}) + \lambda \sum_{j=1}^{k} |\beta_j|, \tag{1.1}$$

with response variable y_i, predictor variables $\boldsymbol{x}_i = (x_{i1}, x_{i2}, \ldots, x_{ik})^\top$, coefficients $\beta_0, \boldsymbol{\beta} = (\beta_1, \beta_2, \ldots, \beta_k)^\top$, sample size n, tuning parameter $\lambda > 0$ and a loss function $d(.)$ to reduce the effect of the residuals. The second term of the target function (1.1) is the lasso penalty term which accounts for variable selection. Among various loss functions, the quantile loss function [10] can be seen as generalization of least-absolute estimation for the prediction of several quantile functions.

Quantile regression methods not only provide a way to estimate quantiles in standard regression models but also significantly expand modeling capabilities by capturing local, quantile-specific dynamics [11]. In comparison to usual linear regression techniques, quantile regression approaches offer several advantages over least/absolute squares-based regression analysis. Whereas least/absolute square methods may exhibit inefficiency when there are highly non-normal errors, quantile methods demonstrate higher robustness to large residuals. This approach often proves to be more flexible compared to least/absolute squares regression models for data modeling. Recently, quantile regression has found various applications in fuzzy regression analysis, e.g., in Hesamian et al. [6], providing a more comprehensive perspective on the relationship between predictors and responses. Combining the lasso method and a uniform-based quantile regression technique [10] then leads to the following target function to be minimized:

$$\sum_{\tau\in(0,1)} \sum_{i=1}^{n} \rho_\tau(y_i - \beta_0 - \boldsymbol{x}_i^\top \boldsymbol{\beta}) + \lambda \sum_{j=1}^{k} |\beta_j|, \tag{1.2}$$

Here, $\rho_\tau(u) = u(\tau - I(u \le 0))$, where I is the indicator function ($\tau \in (0,1)$). Thus, in addition to shrinking some coefficients, the target function (1.2) could improve the predictive accuracy of the linear regression model.

Concerning the integration of fuzzy regression and lasso, there are six techniques presented in previous publications: Hesamian and Akbari [12] applied the common lasso approach in the framework of multiple fuzzy linear regression analysis based on crisp predictors and fuzzy responses. Kong [13] introduced an adaptive lasso approach using least absolute shrinkage for parameter estimation in fuzzy linear regression models. Hesamian et al. [14] enhanced fused lasso estimation for regression analysis based on both fuzzy response and predictor variables. Kashani et al. [15] introduced a penalization estimation method for the coefficients of a fuzzy linear regression model with fuzzy responses and a set of crisp predictors. Kareem and Mohammed [16] proposed a fuzzy bridge regression estimator method to overcome the problem of multicollinearity based on triangular fuzzy responses and crisp predictors. Kim and Jung [17] suggested an α-level estimation algorithm for ridge fuzzy regression modeling, addressing the multicollinearity phenomenon in fuzzy linear regression analysis with fuzzy responses and crisp predictors. Note that most of these models are based on crisp predictors and fuzzy responses, while the approach by Hesamian et al. [14] considers fuzzy responses and fuzzy predictors.

Given that combining quantile regression analysis and lasso could leverage the advantages of both methodologies in efficiently regularizing fuzzy data, we propose a novel fuzzy lasso-quantile-based regression method with both fuzzy predictors and fuzzy responses in this paper. We implement a penalization method for estimating the coefficients of a linear regression model to investigate the dependence of an LR fuzzy response variable on a set of crisp predictors.

The paper is structured as follows. In Section 2, we revisit essential concepts associated with fuzzy numbers. Subsequently, in Section 3 the fuzzy multiple regression model integrated with lasso and quantile methods is introduced. Section 4 elaborates on various application examples. Finally, Section 5 offers concluding remarks.

2. Essentials on fuzzy numbers

This section introduces the main terms for fuzzy sets and fuzzy numbers that are used in this paper. The representation of these basics follows Hesamian et al. [5] and Chukhrova and Johannssen [18].

A fuzzy set is a mapping $\widetilde{A}$ on $\mathbb{X}$ that assigns a degree of membership $0 \le \mu_{\widetilde{A}}(x) \le 1$ to each $x \in \mathbb{X}$. A fuzzy number (**FN**) $\widetilde{A}$ is a convex and normalized fuzzy set on the real line $\mathbb{R}$ with an upper semi-continuous membership function of bounded support. The most often implemented form of **FN**s is given by so called LR-**FN**s. The membership function of an LR-**FN** $\widetilde{A} = (a; l_a, r_a)_{LR}$ is defined by

$$\mu_{\widetilde{A}}(x) = \begin{cases} L\left(\dfrac{a-x}{l_a}\right) & \text{for } x \le a, \\[2ex] R\left(\dfrac{x-a}{r_a}\right) & \text{for } x > a, \end{cases}$$

where $a \in \mathbb{R}$, $l_a > 0$ and $r_a > 0$ are referred to as the center value and the left and right spreads of $\widetilde{A}$, respectively. The shape functions L and R are continuous and strictly decreasing functions from $[0,1]$ to $[0,1]$ satisfying $L(0) = R(0) = 1$ and $L(1) = R(1) = 0$. The set of all LR-**FN**s is represented by $\mathcal{F}_{LR}(\mathbb{R})$. When it holds $L = R$, $\widetilde{A} = (a; l_a, r_a)_{LR}$ is called a symmetric LR-**FN** and is denoted by

$\widetilde{A} = (a; l_a)_L$. The most commonly used special case of an LR-**FN** satisfying $L(x) = R(x) = \max\{0, 1 - x\}$ is known as triangular **FN**. The membership function of a triangular **FN** $\widetilde{A} = (a; l_a, r_a)_T$ is given by:

$$\mu_{\widetilde{A}}(x) = \begin{cases} \frac{x-(a-l_a)}{l_a} & \text{for } a - l_a \le x \le a \\ \frac{a+r_a-x}{r_a} & \text{for } a < x \le a + r_a \\ 0 & \text{otherwise} \end{cases}$$

The addition of two LR-**FN**s $\widetilde{A} = (a; l_a, r_a)_{LR}$ and $\widetilde{B} = (b; l_b, r_b)_{LR}$ is defined as follows:

$$\widetilde{A} \oplus \widetilde{B} = (a + b; l_a + l_b, r_a + r_b)_{LR}$$

3. The quantile-based lasso regression model

We first present the multiple regression model with both responses and predictors being fuzzy, and then we propose parameter estimation using a lasso penalty method, which is adopted with a quantile method (Section 3.1). Next, we introduce four goodness-of-fit criteria and an F-test for comparative analysis between different fuzzy regression models in Sections 3.2 and 3.3, respectively.

3.1. The proposed model and parameter estimation

Let $\{(\widetilde{\boldsymbol{x}}_i, \widetilde{y}_i)\}_{i=1,2,\ldots,n}$ be a fuzzy data set, where $\widetilde{\boldsymbol{x}}_i = (\widetilde{x}_{i1}, \widetilde{x}_{i2}, \ldots, \widetilde{x}_{ik})^\top \in (\mathcal{F}_{LR}(\mathbb{R}))^k$ (fixed values) and $\widetilde{y}_i$ are LR-**FN**s. Consider the problem of approximating $\{(\widetilde{\boldsymbol{x}}_i, \widetilde{y}_i)\}_{i=1,2,\ldots,n}$ by means of the fuzzy linear regression model

$$\widetilde{y}_i = \widetilde{\beta}_0 \oplus (\boldsymbol{x}_i^\top \boldsymbol{\beta}; (\boldsymbol{l}_{\boldsymbol{x}_i})^\top \boldsymbol{l}_{\boldsymbol{\beta}}, (\boldsymbol{r}_{\boldsymbol{x}_i})^\top \boldsymbol{r}_{\boldsymbol{\beta}})_{LR} \oplus \widetilde{\epsilon}_i, \ i = 1, 2, \ldots, n, \tag{3.1}$$

with

1. fuzzy predictors $\widetilde{\boldsymbol{x}}_i = (\boldsymbol{x}_i; \boldsymbol{l}_{\boldsymbol{x}_i}, \boldsymbol{r}_{\boldsymbol{x}_i})_{LR}$, where $\boldsymbol{x}_i = (x_{i1}, x_{i2}, \ldots, x_{ik})^\top$, $\boldsymbol{l}_{\boldsymbol{x}_i} = (l_{x_{i1}}, l_{x_{i2}}, \ldots, l_{x_{ik}})^\top$ and $\boldsymbol{r}_{\boldsymbol{x}_i} = (r_{x_{i1}}, r_{x_{i2}}, \ldots, r_{x_{ik}})^\top$,
2. fuzzy responses $\widetilde{y}_i = (y_i; l_{y_i}, r_{y_i})_{LR}$,
3. an unknown fuzzy intercept $\widetilde{\beta}_0 = (\beta_0; l_{\beta_0}, r_{\beta_0})_{LR}$ with $l_{\beta_0}, r_{\beta_0} > 0$,
4. unknown crisp coefficients $\boldsymbol{\beta} = (\beta_1, \beta_2, \ldots, \beta_k)^\top \in \mathbb{R}^k$, $\boldsymbol{l}_{\boldsymbol{\beta}} = (l_{\beta_1}, l_{\beta_2}, \ldots, l_{\beta_k})^\top \in (\mathbb{R}^+)^k$ and $\boldsymbol{r}_{\boldsymbol{\beta}} = (r_{\beta_1}, r_{\beta_2}, \ldots, r_{\beta_k})^\top \in (\mathbb{R}^+)^k$, and
5. a fuzzy error term $\widetilde{\epsilon}_i = (\epsilon_i; l_{\epsilon_i}, r_{\epsilon_i})_{LR}$ with $l_{\epsilon_i}, r_{\epsilon_i} > 0$.

Following (3.1), we can formulate three separate linear regression models (corresponding to the left spreads, the center and the right spreads of our fuzzy regression model):

(1) $y_i = \beta_0 + \boldsymbol{x}_i^\top \boldsymbol{\beta} + \epsilon_i$, $i = 1, 2, \ldots, n$,
(2) $l_{y_i} = l_{\beta_0} + (\boldsymbol{l}_{\boldsymbol{x}_i})^\top \boldsymbol{l}_{\boldsymbol{\beta}} + l_{\epsilon_i}$, $i = 1, 2, \ldots, n$,
(3) $r_{y_i} = r_{\beta_0} + (\boldsymbol{r}_{\boldsymbol{x}_i})^\top \boldsymbol{r}_{\boldsymbol{\beta}} + r_{\epsilon_i}$, $i = 1, 2, \ldots, n$.

The set of the unknown coefficients is given by $\boldsymbol{\Delta} = \{\boldsymbol{\beta}, \boldsymbol{l}_{\boldsymbol{\beta}}, \boldsymbol{r}_{\boldsymbol{\beta}}, \widetilde{\beta}_0\}$. This set can be estimated using the uniform quantile-based lasso technique on $\tau \in \{0.005, 0.01, \ldots, 0.995\}$. We implement a partitioned subset $\{0.005k : k = 1, 2, \ldots, 99\} = \{0.01, 0.02, \ldots, 0.995\}$ for quantile levels on $[0, 1]$. Our proposed estimation procedure includes estimating the (1) center values $\beta_0, \boldsymbol{\beta}$, (2) left values $l_{\beta_0}, \boldsymbol{l}_\beta$, and (3) right values $r_{\beta_0}, \boldsymbol{r}_{\boldsymbol{\beta}}$ based on a three independent stages as follows:

(1) Estimation of $\beta_0, \boldsymbol{\beta}$:

$$(\widehat{\beta}_0, \widehat{\boldsymbol{\beta}}, \widehat{\lambda}_c) = \arg \min_{\boldsymbol{\beta} \in \mathbb{R}^k, \beta_0 \in \mathbb{R}, \lambda_c > 0} f_c(\boldsymbol{\beta}, \beta_0, \lambda_c),$$

where

$$f_c(\boldsymbol{\beta}, \beta_0, \lambda_c) = \sum_{t=1}^{199} \sum_{i=1}^{n} \rho_{0.005t}(y_i - \beta_0 - \boldsymbol{x}_i^\top \boldsymbol{\beta}) + \lambda_c \sum_{j=1}^{k} |\beta_j|.$$

(2) Estimation of $l_{\beta_0}, \boldsymbol{l}_\beta$:

$$(l_{\widehat{\beta}_0}, \boldsymbol{l}_{\widehat{\boldsymbol{\beta}}}, \widehat{\lambda}_l) = \arg \min_{\boldsymbol{l}_{\boldsymbol{\beta}} \in (\mathbb{R}^k)^+, l_{\beta_0} > 0, \lambda_l > 0} f_l(l_{\beta_0}, \boldsymbol{l}_\beta, \lambda_l),$$

where

$$f_l(l_{\beta_0}, \boldsymbol{l}_\beta, \lambda_l) = \sum_{t=1}^{199} \sum_{i=1}^{n} \rho_{0.005t}(l_{y_i} - l_{\beta_0} - (\boldsymbol{l}_{\boldsymbol{x}_i})^\top \boldsymbol{l}_{\boldsymbol{\beta}}) + \lambda_l \sum_{j=1}^{k} l_{\beta_j}.$$

Table 1
Goodness-of-fit criteria and their interpretations.

Criterion	Range	Interpretation
MSE D^*	$D^* \geq 0$	The closer the value is to zero, the better the model fits the data
MSE δ^2	$\delta^2 \geq 0$	The closer the value is to zero, the better the model fits the data
MARE	MARE $\in [0, \infty)$	Values close to zero indicate a high degree of consistency between fuzzy responses and corresponding predictions
MSM	MSM $\in [0, 1]$	Values close to 1 show a good level of match between the fuzzy responses and their fuzzy predictions

(3) Estimation of $r_{\beta_0}, \boldsymbol{r}_\beta$:

$$(r_{\widehat{\beta}_0}, \boldsymbol{r}_{\widehat{\boldsymbol{\beta}}}, \widehat{\lambda}_r) = \arg \min_{\boldsymbol{r}_\beta \in (\mathbb{R}^k)^+, r_{\beta_0} > 0, \lambda_r > 0} f_r(r_{\beta_0}, \boldsymbol{r}_\beta, \lambda_r),$$

where

$$f_r(r_{\beta_0}, \boldsymbol{r}_\beta, \lambda_r) = \sum_{t=1}^{199} \sum_{i=1}^{n} \rho_{0.005t}(r_{y_i} - r_{\beta_0} - (\boldsymbol{r}_{\boldsymbol{x}_i})^\top \boldsymbol{r}_{\boldsymbol{\beta}}) + \lambda_r \sum_{j=1}^{k} r_{\beta_j}.$$

In the above stages (1)–(3), $\lambda_c, \lambda_l, \lambda_r$ are the tuning parameters. The least local approximation technique is employed to solve the optimization problem by minimizing the target functions. To this end, we implemented a minimization procedure using `Mathematica` in each stage. The model includes $3(k+1)$ regression coefficients in each stage. Therefore, the computational effort of the lasso regularized-quantile-based algorithm can be stated in big-O notation as $O(199(k+1)n)$ for the quantile part and $O(kn)$ for the Shrinkage penalty. Hence, the total complexity of each lasso regularized-quantile-based regression should be $O(199(k+1)n + kn)$ in each stage.

Finally, considering the three-stage procedure, the fuzzy response variable can be predicted in the form of an LR-**FN**:

$$\widetilde{\widehat{y}}_i = \left(\widehat{\beta}_0 + \boldsymbol{x}_i^\top \widehat{\boldsymbol{\beta}}; l_{\widehat{\beta}_0} + (\boldsymbol{l}_{\boldsymbol{x}_i})^\top \boldsymbol{l}_{\widehat{\boldsymbol{\beta}}}, r_{\widehat{\beta}_0} + (\boldsymbol{r}_{\boldsymbol{x}_i})^\top \boldsymbol{r}_{\widehat{\boldsymbol{\beta}}}\right)_{LR}$$

3.2. Goodness-of-fit criteria for comparing two fuzzy regression models

In order to perform comparative analysis, four common goodness-of-fit criteria to evaluate the performance of fuzzy regression models are implemented in this paper:

- Mean Squared Error (MSE) D^*:

$$D^* = \frac{1}{n} \sum_{i=1}^{n} \int_0^1 \left(\left((\widetilde{y}_i)_\alpha^L - (\widetilde{\widehat{y}}_i)_\alpha^L \right)^2 + \left(\left((\widetilde{y}_i)_\alpha^U - (\widetilde{\widehat{y}}_i)_\alpha^U \right)^2 \right) \right) d\alpha.$$

- Mean Squared Error (MSE) δ^2:

$$\delta^2 = \frac{1}{n} \sum_{i=1}^{n} \left((y_i - \widehat{y}_i)^2 + (y_i - l_{y_i} - (\widehat{y}_i - l_{\widehat{y}_i}))^2 + (y_i + r_{y_i} - (\widehat{y}_i + r_{\widehat{y}_i}))^2 \right).$$

- Mean Absolute Relative Error (MARE):

$$\text{MARE} = \frac{1}{n} \sum_{i=1}^{n} \frac{\int_0^1 |\widetilde{y}_i(x) - \widetilde{\widehat{y}}_i(x)| dx}{\int_0^1 \widetilde{y}_i(x) dx}.$$

- Mean Similarity Measure (MSM):

$$\text{MSM} = \frac{1}{n} \sum_{i=1}^{n} \frac{\int \min\{\widetilde{\widehat{y}}_i(x), \widetilde{y}_i(x)\} dx}{\int \max\{\widetilde{\widehat{y}}_i(x), \widetilde{y}_i(x)\} dx}.$$

The interpretation of these criteria can be found in Table 1.

3.3. F-test for comparing two fuzzy regression models

When fitting data using regression models, it may be preferable to have a definitive answer in order to choose the most appropriate model that is not based on the interpretation of goodness-of-fit criteria. Therefore, we introduce an F-test in the following that can be implemented to compare two regression models M_1 and M_2. The test statistics of this test is given by $F = SS_1/SS_2$ with SS_1 as the residual sum of squares for model M_1 and SS_2 as the corresponding sum for model M_2. There are $n - k - 1$ degrees of freedom, where n is the number of data points and k is the number of predictor variables [19]. The F-test is used to test the null hypothesis H_0 : "M_2 has a better fit than M_1" versus the alternative hypothesis H_1 : "M_1 has a better fit than M_2". At significance level α, H_0 can be rejected and H_1 is accepted if $F > F_{\alpha,n-k-1,n-k-1}$.

In the following, we intend to compare two fuzzy regression models $\widetilde{M}_1$ and $\widetilde{M}_2$, that is, we intend to test H_0 : "$\widetilde{M}_2$ has a better fit than $\widetilde{M}_1$" against H_1 : "$\widetilde{M}_1$ has a better fit than $\widetilde{M}_2$". In this regard, note that a fuzzy regression model $\widetilde{M}$ is composed of three ordinary regression models for left spreads, center and right spreads of the fuzzy responses. Therefore, let F_l, F_c, and F_r denote the realized values of the test statistics for comparing the left spreads, the center, and right spreads of the fuzzy regression models $\widetilde{M}_1$ and $\widetilde{M}_2$, respectively. There are then three pairs of hypotheses, which are as follows:

- H_0^l : "the left spreads of $\widetilde{M}_2$ have a better fit than the left spreads of $\widetilde{M}_1$" versus H_1^l : "the left spreads of $\widetilde{M}_1$ have a better fit than the left spreads of $\widetilde{M}_2$".
- H_0^c : "the center of $\widetilde{M}_2$ has a better fit than the center of $\widetilde{M}_1$" versus H_1^c : "the center of $\widetilde{M}_1$ has a better fit than the center of $\widetilde{M}_2$".
- H_0^r : "the right spreads of $\widetilde{M}_2$ have a better fit than the right spreads of $\widetilde{M}_1$" versus H_1^r : "the right spreads of $\widetilde{M}_1$ have a better fit than the right spreads of $\widetilde{M}_2$".

At significance level α, if the values of F_l, F_c, and F_r are larger than $F_{\alpha,n-k-1,n-k-1}$, then the left spreads, the center, and right spreads of $\widetilde{M}_1$ have a better fit than the left spreads, center, and right spreads of $\widetilde{M}_2$. In this case, it is reasonable to state that $\widetilde{M}_1$ has a better fit than $\widetilde{M}_2$. This leads us to define the degree to which $\widetilde{M}_1$ has a better fit than $\widetilde{M}_2$ as follows:

$$D_{\widetilde{M}_1\widetilde{M}_2} = \frac{I(F_c > F_{\alpha,n-k-1,n-k-1}) + I(F_l > F_{\alpha,n-k-1,n-k-1}) + I(F_r > F_{\alpha,n-k-1,n-k-1})}{3},$$

where I represents the indicator function. Note that $D_{\widetilde{M}_1\widetilde{M}_2}$ meets the following properties:

(1) $D_{\widetilde{M}_1\widetilde{M}_2} \in \{0, \frac{1}{3}, \frac{2}{3}, 1\}$,
(2) $D_{\widetilde{M}_1\widetilde{M}_2} = 1 - D_{\widetilde{M}_2\widetilde{M}_1}$,
(3) $D_{\widetilde{M}_1\widetilde{M}_2} = 1$, if and only if the center, left and right spreads of $\widetilde{M}_1$ have a better fit than $\widetilde{M}_2$.
(4) $D_{\widetilde{M}_1\widetilde{M}_2} = 0$, if and only if the center, left and right spreads of $\widetilde{M}_2$ have a better fit than $\widetilde{M}_1$.

According to property (2), it is reasonable to state that (at significance level α) $\widetilde{M}_1$ has a better fit than $\widetilde{M}_2$ if $D_{\widetilde{M}_1\widetilde{M}_2} > D_{\widetilde{M}_2\widetilde{M}_1}$, that is, when it holds $D_{\widetilde{M}_1\widetilde{M}_2} > 0.5$.

4. Numerical examples

We compare the fuzzy quantile-based lasso regression model with fuzzy regression models established in the recent past. Our aim is to assess the effectiveness of our methodology through examples that are both simulation studies and real-data applications. For the comparative analysis, we select competitors that satisfy two criteria: (1) comparability, meaning they are also founded on both fuzzy response and fuzzy predictor variables, and (2) superiority demonstrated in prior applications.

Example 4.1. This example is based on simulations, where we consider 10 artificial fuzzy data sets (with $n = 1000$ in each case) and the following model:

$$\widetilde{y}_i = \widetilde{\beta}_0 \oplus (\boldsymbol{x}_i^\top \boldsymbol{\beta}; (\boldsymbol{l}_{\boldsymbol{x}_i})^\top \boldsymbol{l}_{\boldsymbol{\beta}}, (\boldsymbol{r}_{\boldsymbol{x}_i})^\top \boldsymbol{r}_{\boldsymbol{\beta}})_{LR} \oplus \widetilde{\epsilon}_i,\ i = 1, 2, \ldots, 1000$$

It holds $k = 3$ and

1. $\widetilde{x}_{i1} = (x_{i1}; l_{x_{i1}}, r_{x_{i1}})_{LR}$ with $x_{i1} = z_1 + e_i$, $z_1 \sim N(0, 0.1)$, $e_i \sim N(0, 0.05)$,
   $\widetilde{x}_{i2} = (x_{i2}; l_{x_{i2}}, r_{x_{i2}})_{LR}$ with $x_{i2} = z_2 + e_i$, $z_2 \sim N(0, 0.05)$, $e_i \sim N(0, 0.10)$,
   $\widetilde{x}_{i3} = (x_{i3}; l_{x_{i3}}, r_{x_{i3}})_{LR}$ with $x_{i3} = z_3 + e_i$, $z_3 \sim N(0, 0.15)$, $e_i \sim N(0, 0.15)$,
2. $l_{x_{ij}} \sim U(0.01, 0.03)$ and $r_{x_{ij}} \sim U(0.02, 0.04)$, $j = 1, 2, 3$,
3. $\widetilde{\beta}_0 = (0; 0.4, 0.6)_{LR}$, $\{\beta_1, \beta_2, \beta_3\} = \{2.3, -1.2, 0.8\}$, $\{l_{\beta_1}, l_{\beta_2}, l_{\beta_3}\} = \{0.5, 0.5, 0.5\}$ and $\{r_{\beta_1}, r_{\beta_2}, r_{\beta_3}\} = \{1, 1, 1\}$,
4. $\widetilde{\epsilon}_i = (\epsilon_i; l_{\epsilon_i}, r_{\epsilon_i})_T$ with $l_{\epsilon_i} = 0.2|\epsilon_i|$, $r_{\epsilon_i} = 0.15|\epsilon_i|$ and $\epsilon_i \sim N(0, 1.5)$,
5. $L(x) = 1 - x^2$ and $R(x) = \sqrt{1 - x^2}$.

Upon implementing the introduced quantile-based lasso method, we derived the average values $\overline{D}^* = 0.0136$, $\overline{\delta}^2 = 0.0312$, $\overline{\text{MARE}} = 0.27$ and $\overline{\text{MSM}} = 0.86$ for the considered goodness-of-fit criteria across the 10 simulated data sets. Additionally, the performance results for the 6th data set are exemplified in Table 2. Based on the obtained values of the goodness-of-fit measures (average values and results in Table 2), we can conclude that the fuzzy quantile-based lasso regression model yields low values of D^*, δ^2, and MARE. The closeness of the δ^2 and D^* values to zero shows a good fit of the model to the data, while the observed MARE and MSM values indicate a strong alignment between the fuzzy responses and their predicted values.

Note that apart from Hesamian et al. [14], comparisons with other methods are not feasible since they primarily rely on triangular **FN**s or symmetric **FN**s. Hence, to assess the performance of our proposed model, we compare it with the method from Hesamian et al.

Table 2
Comparison of results for the 6th simulated data set in Example 4.1.

Method	Estimated model $\widetilde{y}_i = (\widehat{y}_i; l_{\widehat{y}_i}, r_{\widehat{y}_i})_{LR}$	D^*	δ^2	MARE	MSM
$\widetilde{M}_1$: Proposed model	$\widehat{y}_i = -0.0058 - 0.0184x_{i1} + 0.0024x_{i2} + 0.8270x_{i3}$, $l_{\widehat{y}_i} = 0.4099 + 0.6537l_{x_{i3}}$, $r_{\widehat{y}_i} = 0.6052 + 0.0276r_{x_{i1}} + 0.0669r_{x_{i2}} + 1.0563r_{x_{i3}}$.	0.0114	0.0323	0.26	0.87
$\widetilde{M}_2$: Hesamian et al. [14]	$\widehat{y}_i = 0.0056 + 0.0039x_{i1} + 0.0079x_{i2} + 0.7897x_{i3}$, $l_{\widehat{y}_i} = 0.4143 + 0.1520l_{x_{i1}} + 0.1520l_{x_{i2}} + 0.1523l_{x_{i3}}$, $r_{\widehat{y}_i} = 0.6052 + 0.1206r_{x_{i1}} + 0.0392r_{x_{i2}} + 1.0041r_{x_{i3}}$.	0.0227	0.0442	0.29	0.79

Table 3
Testing H_0 : "$\widetilde{M}_2$ has a better fit than $\widetilde{M}_1$" versus H_1 : "$\widetilde{M}_1$ has a better fit than $\widetilde{M}_2$" in Example 4.1 at significance level $\alpha = 0.05$.

Model	$\overline{SS}_c$	$\overline{SS}_l$	$\overline{SS}_r$	$\overline{F}_c$	$\overline{F}_l$	$\overline{F}_r$	$\overline{D}_{\widetilde{M}_1\widetilde{M}_2}$
$\widetilde{M}_1$: Proposed model	9.083	1.153	0.086	–	–	–	–
$\widetilde{M}_2$: Hesamian et al. [14]	10.261	1.844	0.101	1.13	1.60	1.17	1

[14]. In this regard, the average values of the goodness-of-fit criteria are $\overline{D}^* = 0.0458$, $\overline{\delta}^2 = 0.0632$, $\overline{\text{MARE}} = 0.58$, and $\overline{\text{MSM}} = 0.80$. The results of these criteria for the 6th simulated data set for the method by Hesamian et al. [14] are also given in Table 2. When comparing the realizations of the performance indices, one observes slightly lower values of D^*, δ^2, MARE as well as a higher value of MSM for our model.

In addition to comparing the goodness-of-fit criteria, the results of the F-test for testing H_0 : "$\widetilde{M}_2$ has a better fit than $\widetilde{M}_1$" versus H_1 : "$\widetilde{M}_1$ has a better fit than $\widetilde{M}_2$", where $\widetilde{M}_1$ is our proposed model and $\widetilde{M}_2$ is the model introduced by Hesamian et al. [14], are given in Table 3. The average values of the F-statistics for left spreads, center and right spreads for our model and for the model of Hesamian et al. [14] across all simulated data sets are also listed in Table 3. Based on these results, we observe $\overline{D}_{\widetilde{M}_1\widetilde{M}_2} > 0.5$ (with $F_{0.05,996,996} = 1.109$) so that we can reject H_0, and therefore, we can conclude that our proposed model is to prefer.

Example 4.2. Let us consider the real data set discussed in D'Urso [20] that consists of two fuzzy predictors (decision on cooking $\widetilde{x}_1$, decision on environment $\widetilde{x}_2$) and a fuzzy response (decision on cellar $\widetilde{y}$). In this example, the following fuzzy multiple linear regression model is utilized:

$$\widetilde{y}_i = (\beta_0; l_{\beta_0}, r_{\beta_0})_{LR} \oplus \left(\sum_{j=1}^{2} x_{ij}\beta_j; \sum_{j=1}^{2} l_{x_{ij}} l_{\beta_j}, \sum_{j=1}^{2} r_{x_{ij}} r_{\beta_j}\right)_{LR} \oplus \widetilde{\epsilon}_i,\ \ i = 1, \dots, 30.$$

According to the proposed method, the unknown parameters of the model can be estimated as follows:

$$(\widehat{\beta}_0, \widehat{\beta}_1, \widehat{\beta}_2, \widehat{\lambda}_c) = \arg \min_{\beta_0,\beta_1,\beta_2,\lambda_c>0} \left\{ \sum_{t=1}^{199}\sum_{i=1}^{30} \rho_{0.005t}(y_i - \beta_0 - \sum_{j=1}^{2} x_{ij}\beta_j) + \lambda_c \sum_{j=1}^{2} |\beta_j| \right\},$$

$$(l_{\widehat{\beta}_0}, l_{\widehat{\beta}_1}, l_{\widehat{\beta}_2}, \widehat{\lambda}_l) = \arg \min_{l_{\beta_0}>0, l_{\beta_1}>0, l_{\beta_2}>0, \lambda_l>0} \left\{ \sum_{t=1}^{199}\sum_{i=1}^{30} \rho_{0.005t}(l_{y_i} - l_{\beta_0} - \sum_{j=1}^{2} l_{x_{ij}} l_{\beta_j}) + \lambda_l \sum_{j=1}^{2} l_{\beta_j} \right\},$$

$$(r_{\widehat{\beta}_0}, r_{\widehat{\beta}_1}, r_{\widehat{\beta}_2}, \widehat{\lambda}_r) = \arg \min_{r_{\beta_0}>0, r_{\beta_1}>0, r_{\beta_2}, \lambda_r>0} \left\{ \sum_{t=1}^{199}\sum_{i=1}^{30} \rho_{0.005t}(r_{y_i} - r_{\beta_0} - \sum_{j=1}^{2} r_{x_{ij}} r_{\beta_j}) + \lambda_r \sum_{j=1}^{2} r_{\beta_j} \right\}.$$

To evaluate the performance of this model, we compare our model with the methods from D'Urso [20], Lu and Wang [21], Chachi and Taheri [22] and Hesamian et al. [14]. The results of these models (estimated parameters and goodness-of-fit criteria) can be found in Table 4. Comparing the realizations of the goodness-of-fit criteria, we observe lower values of D^*, δ^2, MARE, and a higher value of MSM for our introduced model. In addition, the results of the F-test are given in Table 5. It can be seen that it holds $D_{\widetilde{M}_1\widetilde{M}_2} > 0.5$ (with $F_{0.05,27,27} = 1.90$) for all specified models which reveals that the proposed model is better suited than the other models. Thus, the proposed fuzzy quantile-based lasso regression model shows a better performance compared to the other four fuzzy regression models for the considered data set.

Example 4.3. Let us consider a fuzzy input–output data set taken from Chachi and Taheri [22]. There are eight pairs of symmetric triangular **FN**s in this data set. We suggest a fuzzy univariate linear regression model:

$$\widetilde{y}_i = (\beta_0; l_{\beta_0}, r_{\beta_0})_{LR} \oplus (x_i\beta_j; l_{x_i} l_{\beta_1}, r_{x_i} r_{\beta_1})_{LR} \oplus \widetilde{\epsilon}_i,\ \ i = 1, \dots, 8.$$

We implement the introduced procedure to estimate the model parameters as follows:

$$(\widehat{\beta}_0, \widehat{\beta}_1, \widehat{\lambda}_c) = \arg \min_{\beta_1,\beta_0,\lambda_c>0} \left\{ \sum_{t=1}^{199}\sum_{i=1}^{8} \rho_{0.005t}(y_i - \beta_0 - x_i\beta_1) + \lambda_c |\beta_1| \right\},$$

Table 4
The performance of various models for the restaurant data set in Example 4.2.

Method	Estimated model $\widehat{\tilde{y}}_i = (\hat{y}_i, l_{\hat{y}_i}, r_{\hat{y}_i})_T$	Performance
D'Urso [20]	$\hat{y}_i = (1, x_{i1}, x_{i2})^\top \boldsymbol{x} + (1, l_{x_{i1}}, l_{x_{i2}})^\top \boldsymbol{l_x} + (1, r_{x_{i1}}, r_{x_{i2}})^\top \boldsymbol{r_x}$ $\boldsymbol{x} = (0.6498, 0.4542, 0.4924)^\top$ $\boldsymbol{l_x} = (-1.8685, 2.3604, 0.7392)^\top$ $\boldsymbol{r_x} = (-0.2333, -0.1339, 0.1271)^\top$ $l_{\hat{y}_i} = -0.401173 + 0.1173197\,\hat{y}_i$ $r_{\hat{y}_i} = -0.650102 + 0.2306911\,\hat{y}_i$	$D^* = 1.08$, $\delta^2 = 2.45$, MARE = 1.34, MSM = 0.22.
Lu and Wang [21]	$\hat{y}_i = -1.66 + 1.33\,x_{i1}$ $l_{\hat{y}_i} = 0.8 - 0.91\,l_{x_{i1}} - 0.65\,l_{x_{i2}} + 0.63\,x_{i1}$ $-0.28\,x_{i2} - 1.48\,r_{x_{i2}}$ $r_{\hat{y}_i} = 0.08 + 0.37\,l_{x_{i1}} + 0.09\,l_{x_{i2}} - 0.12\,x_{i1}$ $+0.03\,x_{i2} + r_{x_{i1}} + 0.2\,r_{x_{i2}}$	$D^* = 2.022$, $\delta^2 = 2.15$, MARE = 0.90, MSM = 0.56.
Chachi and Taheri [22]	$\hat{y}_i = -1.7638 + 0.7864\,x_{i1} + 0.4245\,x_{i2}$ $l_{\hat{y}_i} = 0.7864\,l_{x_{i1}} + 0.4245\,l_{x_{i2}}$ $r_{\hat{y}_i} = 0.7864\,r_{x_{i1}} + 0.4245\,r_{x_{i2}}$	$D^* = 1.23$, $\delta^2 = 2.002$, MARE = 1.11, MSM = 0.38.
Hesamian et al. [14]	$\hat{y}_i = -0.2958 + 1.0183x_{i1} + 0.0209x_{i2}$ $l_{\hat{y}_i} = l_{x_{i1}}$ $r_{\hat{y}_i} = 0.0930 + 0.8139r_{x_{i1}} + 4.9826 \times 10^{-7} r_{x_{i2}}$	$D^* = 1.36$, $\delta^2 = 2.11$, MARE = 1.15, MSM = 0.49.
Proposed model	$\hat{y}_i = 6.7640 + 0.0035x_{i1} + 0.0296x_{i2}$ $l_{\hat{y}_i} = 0.5 + 2.5714 \times 10^{-8} l_{x_{i1}} + 5.4927 \times 10^{-8} l_{x_{i2}}$ $r_{\hat{y}_i} = 0.7732 + 0.21398r_{x_{i1}} + 0.2101r_{x_{i2}}$	$D^* = 0.50$, $\delta^2 = 0.81$, MARE = 0.38, MSM = 0.73.

Table 5
Testing H_0 : "$\widetilde{M}_2$ has a better fit than $\widetilde{M}_1$" versus H_1 : "$\widetilde{M}_1$ has a better fit than $\widetilde{M}_2$" in Example 4.2 at significance level $\alpha = 0.05$.

Model	SS_c	SS_l	SS_r	F_c	F_l	F_r	$D_{\widetilde{M}_1\widetilde{M}_2}$
$\widetilde{M}_1$: Proposed model	6.45408	0.4375	2.46034	–	–	–	–
$\widetilde{M}_2$: D'Urso [20]	16.0846	1.48426	2.05761	2.49216	3.39258	0.83631	0.667
$\widetilde{M}_2$: Lu and Wang [21]	30.8303	6.6664	2.70398	4.77687	15.2376	1.09903	0.667
$\widetilde{M}_2$: Chachi and Taheri [22]	17.2174	2.50546	7.1098	2.66767	5.72677	2.88976	1
$\widetilde{M}_2$: Hesamian et al. [14]	19.6499	1.75000	2.43013	3.04457	4.00000	0.987719	0.667

Table 6
Fuzzy predicted values in Example 4.3.

No.	$\widehat{\tilde{y}}$	$\tilde{y}$
1	$(4.832; 0.499)_T$	$(4.0; 0.5)_T$
2	$(5.499; 0.500)_T$	$(5.5; 0.5)_T$
3	$(6.388; 0.999)_T$	$(7.5; 1.0)_T$
4	$(7.055; 0.499)_T$	$(6.5; 0.5)_T$
5	$(7.721; 0.500)_T$	$(8.5; 0.5)_T$
6	$(8.610; 1.000)_T$	$(8.0; 1.0)_T$
7	$(8.833; 0.499)_T$	$(10.5; 0.5)_T$
8	$(9.500; 0.500)_T$	$(9.5; 0.5)_T$

$$(l_{\hat{\beta}_0}, l_{\hat{\beta}_1}, \hat{\lambda}_l) = \arg \min_{l_{\beta_1}>0, l_{\beta_0}>0, \lambda_l>0} \left\{ \sum_{t=1}^{199} \sum_{i=1}^{8} \rho_{0.005t}(l_{y_i} - l_{\beta_0} - l_{x_i} l_{\beta_1}) + \lambda_l l_{\beta_1} \right\},$$

$$(r_{\hat{\beta}_0}, r_{\hat{\beta}_1}, \hat{\lambda}_r) = \arg \min_{r_{\beta_1}>0, r_{\beta_0}>0, \lambda_r>0} \left\{ \sum_{t=1}^{199} \sum_{i=1}^{8} \rho_{0.005t}(r_{y_i} - r_{\beta_0} - r_{x_i} r_{\beta_1}) + \lambda_r r_{\beta_1} \right\}.$$

First, Table 6 presents the fuzzy predicted values generated by the quantile-based lasso regression model. Second, Table 7 provides the estimated coefficients and performance metrics for the fuzzy regression models under consideration. Comparing the results of D^*, δ^2, MARE and MSM across the considered models, it is obvious that the lowest values for D^*, δ^2, MARE and the highest value for MSM have been obtained for the fuzzy quantile-based lasso regression model. Third, the results of the F-test are presented in Table 8. Note that the method of Nasrabadi and Nasrabadi [23] can only evaluate the center of the regression model, and thus does not provide a fuzzy regression model for our comparison based on the F-test. Table 8, it can be seen that it holds $D_{\widetilde{M}_1\widetilde{M}_2} > 0.5$ (with $F_{0.05,6,6} = 4.28$) for all specified models which reveals that the proposed model has a better fit than the other models for the considered data set.

Table 7
The performance of various models for the data set in Example 4.3.

Method	Estimated model	D^*	δ^2	MARE	MSM
Kao and Chyu [24]	$\widehat{\widetilde{y}}_i = (3.5724 + 0.5193\, x_i, 0.24 + 0.5193\, l_{x_i})_T$	10.36	15.66	1.50	0.15
Kao and Chyu [25]	$\widehat{\widetilde{y}}_i = (3.3936 + 0.543\, x_i, 0.9644 + 0.543 l_{x_i})_T$	12.90	23.12	2.02	0.23
Nasrabadi and Nasrabadi [23]	$\widehat{\widetilde{y}}_i = (3.5767 + 0.5467\, x_i, l_{x_i})_T$	11.15	16.72	1.49	0.20
Bargiela et al. [26]	$\widehat{\widetilde{y}}_i = (3.446 + 0.536\, x_i, 0.536\, l_{x_i})_T$	10.84	17.01	1.34	0.07
Chen and Dang [27]	$\widehat{\widetilde{y}}_i = 3.5284 + 0.5298\, \widetilde{x}_i + \widetilde{E}_i$, $\widetilde{E}_1 = (0; 0.234)_T$, $\widetilde{E}_2 = (0; 0.234)_T$, $\widetilde{E}_3 = (0; 0.935)_T$, $\widetilde{E}_4 = (0; 0.234)_T$, $\widetilde{E}_5 = (0; 0.234)_T$, $\widetilde{E}_6 = (0; 0)_T$, $\widetilde{E}_7 = (0; 0.234)_T$, $\widetilde{E}_8 = (0; 0.234)_T$	10.61	16.36	1.53	0.15
Chachi and Taheri [22]	$\widehat{\widetilde{y}}_i = (3.530 + 0.525\, x_i, 0.525 l_{x_i})_T$	10.82	17.02	1.33	0.08
Hesamian et al. [14]	$\widehat{\widetilde{y}}_i = (3.577 + 0.4739 x_i, 4.7728 \times 10^{-6} l_{x_i})_T$	1.56	2.35	1.30	0.27
Proposed model	$\widehat{\widetilde{y}}_i = (3.9431 + 0.4445 x_i; 4.8364 \times 10^{-8} + l_{x_i})_T$	1.49	2.25	1.26	0.32

Table 8
Testing H_0 : "$\widetilde{M}_2$ has a better fit than $\widetilde{M}_1$" versus H_1 : "$\widetilde{M}_1$ has a better fit than $\widetilde{M}_2$" in Example 4.3 at significance level $\alpha = 0.05$.

Model	SS_c	SS_l	F_c	F_l	$D_{\widetilde{M}_1 \widetilde{M}_2}$
$\widetilde{M}_1$: Proposed model	5.99590	4.70828×10^{-15}	–	–	–
$\widetilde{M}_2$: Kao and Chyu [24]	5.14311	0.11587	0.85808	2.46106×10^{13}	0.667
$\widetilde{M}_2$: Kao and Chyu [25]	5.19778	3.7642	0.86719	7.99486×10^{14}	0.667
$\widetilde{M}_2$: Nasrabadi and Nasrabadi [23]	5.57279	0	0.92977	0	0
$\widetilde{M}_2$: Bargiela et al. [26]	5.17021	0.75353	0.86260	1.60045×10^{14}	0.667
$\widetilde{M}_2$: Chen and Dang [27]	5.16359	99.9493	0.86149	2.12284×10^{16}	0.667
$\widetilde{M}_2$: Chachi and Taheri [22]	5.14623	0.78968	0.86860	1.67723×10^{14}	0.667
$\widetilde{M}_2$: Hesamian et al. [14]	6.26851	4.56888×10^{-11}	1.04584	9703.93000	0.667

5. Conclusions

Lasso is a powerful tool for multiple regression models to improve predictive performance, enable feature selection and ensure model interpretability — all while maintaining computational efficiency. Quantile regression methods extend classical least squares estimation to encompass an ensemble of models for various conditional quantile functions, offering flexibility in handling extreme residuals. In this article, by combining lasso and quantile regression, we introduced the lasso regularized quantile-based regression model with fuzzy predictors and fuzzy responses, employing the sum of absolute coefficients as the penalty. We utilized a unified quantile loss function to estimate the model parameters. Four goodness-of-fit criteria were employed to assess the performance of the introduced model, comparing it with other popular fuzzy regression models. Through a simulation study, we demonstrated the benefits of our method over existing fuzzy regression models. Furthermore, numerical evaluations showcased its enhanced efficiency compared to prior applications.

One advantage of the proposed method is its ability to consider all types of LR fuzzy numbers for the fuzzy responses and fuzzy predictors. Since the proposed method requires the estimation of unknown fuzzy coefficients in three different stages, it reduces the computational time necessary to estimate the parameters compared to other methods (based on triangular fuzzy responses). Therefore, the proposed method can be easily applied to fuzzy linear regression with large data sets.

However, the proposed methodology can only be applied to fuzzy linear regression models. Therefore, extending the proposed methodology to nonlinear regression models with fuzzy predictors and responses is a potential topic for future studies. Further, as observed in conventional regression analysis, the lasso regularized quantile-based regression method may exhibit drawbacks such as overfitting and optimism bias, particularly when influential predictors heavily impact the response variable. Thus, exploring extensions of the fuzzy regression model using alternative shrinkage methods for fuzzy data presents a promising avenue for future research. Additionally, examining the sensitivity of the fuzzy regression method presented to outliers could provide valuable insights for further analysis.

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Data availability

The data supporting the findings of this study are available in the respective references, as mentioned in the main text.

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